T10 (old label T11): feasible-addition hazards and ordered paths

Proof, version 4

Lead: Claude Opus 5.5 (high). Status: not adopted.

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The statement proved is `STATEMENT.md` in this folder; all notation is as defined there (it follows `00_COMMON.md`).

1 Cited results

All from wave 1, `proofs/COLLECTED/COLLECTED_PROOFS_v2.tex`.

- **T09(ii).** At temperature φ the Gibbs-j kernel is

$$K_{G,j}(S, S^{\pm}) = \frac{\gamma_\varphi(S^{\pm})}{\gamma_\varphi(S^+) + \gamma_\varphi(S^-)}, \qquad S^+ = S \cup \{j\},\ S^- = S \setminus \{j\}, \tag{1}$$

  a Markov kernel on Ω, with $\gamma_\varphi = 0$ off Ω, so an addition at $|S| = K$ has probability 0. By definition $o_j(S, \varphi) = K_{G,j}(S, S^+)$. T09(ii) also states that every elementary kernel (Gibbs, uniform swap, neighbour swap, independence) is a q_φ-reversible Markov kernel on Ω for $\varphi \in [0, 1]$; this is used only in Remark 6(c).

- **T22, Lemma 1 (positivity and a closed form).** For every $S \subseteq [p]$: $Q_S \geq 0$, $D_S \geq 1$ and $m(S) \in (0, \infty)$; every $r_b(U)$, $w(k)$ $(0 \leq k \leq p)$ and $\gamma_\varphi(S)$ $(S \in \Omega)$ is positive and finite.

- **T22, Lemma 2 (one addition).** For every $T \subseteq [p]$ (no cap; any design, singular G_T included) and $j \notin T$: $Q_{T\cup\{j\}} \leq Q_T$, $D_{T\cup\{j\}}/D_T \leq n + 1$, and hence

$$\frac{m(T \cup \{j\})}{m(T)} \geq (n + 1)^{-1/2}. \tag{2}$$

- **T22, Lemma 3 (prior ratios).** For $0 \leq k \leq p - 1$, $w(k + 1)/w(k) = (k + 1)/(2p - k - 1)$.

The three T22 lemmas are deterministic facts about the score formula of 00_COMMON §1 and use none of the hypotheses of T22's theorem. (For the reader: Q_T is the minimum over $\beta \in \mathbb{R}^T$ of $\|y - X_T\beta\|^2 + n^{-1}(\|X_T\beta\|^2 + \|\beta\|^2)$, and padding with $\beta_j = 0$ embeds the T-problem in the $T \cup \{j\}$-problem, which gives $Q_{T\cup\{j\}} \leq Q_T$; and with $M_T = X_T X_T^\top$, Sylvester's identity and the matrix determinant lemma give $D_{T\cup\{j\}}/D_T = (1 + (n + 1)\xi)/(1 + \zeta) \leq n + 1$, where $\xi = x_j^\top (I_n + (n + 1)M_T)^{-1} x_j \leq \zeta = x_j^\top (I_n + M_T)^{-1} x_j$. Then $m(T')/m(T) = (D_{T'}/D_T)^{-1/2}\big((2b_0 + Q_{T'})/(2b_0 + Q_T)\big)^{-\nu}$ with $\nu > 0$.)

2 Part (i)

Fix $S \in \Omega$ with $k = |S| < K$, $j \notin S$ and $\varphi \in [0,1]$. Then $|S+j| = k+1 \le K$, so $S+j \in \Omega$, and $w(k+1)/w(k) = \binom{2p-1}{k}/\binom{2p-1}{k+1} = (k+1)/(2p-k-1)$, by T22 Lemma 3, since $k \le K-1 \le p-1$ (so $2p-k-1 \ge p \ge 1$).

By T22 Lemma 1, $\varpi(S)$, every $r_b(\cdot)$, $\mathrm{BF}(\cdot)$ and $\gamma_\varphi(\cdot)$ are positive and finite on Ω, so the ratios and real powers below are well defined. Since $\gamma_\varphi = \gamma_0 e^{\varphi d}$ and $\gamma_0(S)e^{d(S)} = \varpi(S)\mathrm{BF}(S)$,

$$\gamma_\varphi(S) = \varpi(S)\Big[\prod_b r_b(S \cap C_b)\Big]^{1-\varphi}\mathrm{BF}(S)^\varphi \qquad (S \in \Omega). \tag{3}$$

Let b be the block containing j and $U = S \cap C_b$. Then $(S+j)\cap C_b = U+j$ and $(S+j)\cap C_{b'} = S \cap C_{b'}$ for $b' \neq b$. By (3), with $r_b = \mathrm{BF} = m/m(\emptyset)$ on subsets of C_b,

$$a_j(S,\varphi) = \frac{\gamma_\varphi(S+j)}{\gamma_\varphi(S)} = \frac{w(k+1)}{w(k)}\Big[\frac{m(U+j)}{m(U)}\Big]^{1-\varphi}\Big[\frac{m(S+j)}{m(S)}\Big]^{\varphi}$$

(W_K cancels because both sets lie in Ω). Both bracketed ratios are $\ge (n+1)^{-1/2}$ by (2) (with $T = U$ and $T = S$; $j \notin U \subseteq S$), and the exponents $1-\varphi, \varphi$ are nonnegative with sum 1. Hence

$$a_j(S,\varphi) \ge \frac{k+1}{2p-k-1}\,(n+1)^{-(1-\varphi)/2}(n+1)^{-\varphi/2} = r_k.$$

Since $j \notin S$, $S^- = S$ and $S^+ = S+j$ in (1), so $o_j(S,\varphi) = a_j/(1+a_j)$; $x \mapsto x/(1+x)$ is increasing on $[0,\infty)$, so $o_j(S,\varphi) \ge r_k/(1+r_k) = h_k$. Finally $\pi_0 = \gamma_0/\bar{Z}_0$ and $\gamma_0 = \gamma_\varphi|_{\varphi=0}$, so $\pi_0(S+j)/\pi_0(S) = a_j(S,0) \ge r_k$ (this clause is also T22's lemma "One feasible addition under π_0"). □

Remark 1. Judging by the sketch in Section 1 (not by the cited statements alone), the argument for part (i) uses no property of the standardised data; it would hold for any real X and $y \neq 0$. This remark is not part of the statement and is not used.

3 Part (ii): the ordered-path lower bound

Fix S, J, a, φ as in (ii). If $a = 0$ the bound is 1 and $X_0 = S = S \cup J$. Let $a \ge 1$. A round draws σ uniformly from the $m!$ orders, independently of the state, and then $X_i \sim K_{\sigma(i)}(X_{i-1}, \cdot)$ with fresh randomness (EX-S); thus for every order σ_0 and states $x_1, \dots, x_m$,

$$\mathbb{P}(\sigma = \sigma_0, X_1 = x_1, \dots, X_m = x_m \mid X_0 = S) = \frac{1}{m!}\prod_{i=1}^{m} K_{\sigma_0(i)}(x_{i-1}, x_i), \qquad x_0 = S. \tag{4}$$

Let $\mathcal{E}$ be the set of orders whose first a labels are the a Gibbs labels of the coordinates of J, in any order. There are $a!\,(m-a)!$ such orders ($a \le K-k \le p \le m$), so $\mathbb{P}(\sigma \in \mathcal{E}) = \binom{m}{a}^{-1}$.

Fix $\sigma_0 \in \mathcal{E}$ and let $j_1, \dots, j_a$ be the coordinates of its first a labels, in order; $\{j_1, \dots, j_a\} = J$. Put $S_0 = S$ and $S_i = S \cup \{j_1, \dots, j_i\}$. For $1 \le i \le a$: $S_{i-1} \in \Omega$, $|S_{i-1}| = k+i-1 \le k+a-1 < K$ and $j_i \notin S_{i-1}$, so by (1) and part (i), $K_{G,j_i}(S_{i-1}, S_i) = o_{j_i}(S_{i-1}, \varphi) \ge h_{k+i-1}$. Summing (4) over $x_{a+1}, \dots, x_m$ (each K is a Markov kernel),

$$\mathbb{P}(\sigma = \sigma_0,\ X_i = S_i\ (1 \le i \le a) \mid X_0 = S) = \frac{1}{m!}\prod_{i=1}^{a} o_{j_i}(S_{i-1}, \varphi) \ge \frac{1}{m!}\prod_{i=0}^{a-1} h_{k+i}.$$

On this event $X_a = S \cup J$, so $S \cup J$ is visited. Summing over the $a!(m-a)!$ orders $\sigma_0 \in \mathcal{E}$ (disjoint events),

$$\mathbb{P}(\text{visit } S\cup J \mid X_0 = S) \;\geq\; \mathbb{P}\big(\sigma \in \mathcal{E} \text{ and each of the first } a \text{ Gibbs steps adds} \mid X_0 = S\big) \;\geq\; \binom{m}{a}^{-1} \prod_{i=0}^{a-1} h_{k+i}.$$

With $a = 1$, $J = \{j\}$: $\mathbb{P}(\text{visit } S + j \mid X_0 = S) \geq h_k/m$. □

4 A two-variable example: parts (ii-neg) and (iv-a)

Data. Let $n \geq 400$ and $p = 2$. Since $\{v \in \mathbb{R}^n : \mathbf{1}^\top v = 0\}$ has dimension $n - 1 \geq 3$, there are mutually orthogonal centred vectors x_1, x_2, y with $\|x_1\|^2 = \|x_2\|^2 = \|y\|^2 = n - 1$. Take $X^r = [x_1\ x_2]$, $y^r = y$. These raw data satisfy the data conditions quoted in STATEMENT.md from 00_COMMON §1 ($n \geq 3$, finite entries, no constant column, y^r non-constant), and they are already standardised: $Cx_j = x_j$, the sample SD with divisor $n - 1$ is $\sqrt{\|x_j\|^2/(n-1)} = 1$, so $D_x = I$ and $X = CX^r D_x^{-1} = X^r$; likewise $s_y = 1$ and $y = Cy^r/s_y = y^r$. Put $\beta := n^{-1/2} \in (0, 1/20]$.

Method objects. $K = \min(2, \lfloor n/2 \rfloor) = 2$, so Ω consists of the four sets $\emptyset$, $\mathsf{a} = \{1\}$, $\mathsf{b} = \{2\}$, $\mathsf{ab} = \{1, 2\}$. The block-size bound is $\min(\lfloor\sqrt{n}\rfloor, \lfloor\sqrt{2}\rfloor, 16) = 1$, so the blocks are $\{1\}$ and $\{2\}$, there is one neighbour swap, and $m = p+1+2 = 5$. The labels are G_1, G_2 (Gibbs), U (uniform swap), N (neighbour swap) and I (independence). The neighbour rule always includes j's single largest-$|r|$ partner, which for $p = 2$ is the other variable; so $N(1) = \{2\}$ and $N(2) = \{1\}$.

Scores. For $T \subseteq \{1, 2\}$ with $|T| = t$: $G_T = (n-1)I_t$ and $z_T = 0$, so $Q_T = y^\top y$, $H_T = ((n+1)(n-1)+1)I_t = n^2 I_t$, $G_T + I_t = nI_t$, $D_T = n^{2t}/n^t = n^t$ and $\mathrm{BF}(T) = n^{-t/2} = \beta^t$. The tables are $r_{\{j\}}(\{j\}) = \beta$, so $\prod_b r_b(S \cap C_b) = \beta^{|S|} = \mathrm{BF}(S)$ and $d \equiv 0$. Hence for every $\varphi \in [0, 1]$, $\gamma_\varphi(S) = \gamma_0(S) = \varpi(S)\beta^{|S|}$, and with $w(0) = \frac{1}{2}$, $w(1) = w(2) = \frac{1}{6}$,

$$\gamma_\varphi \propto \left(\tfrac{1}{2},\ \tfrac{\beta}{6},\ \tfrac{\beta}{6},\ \tfrac{\beta^2}{6}\right) \text{ on } (\emptyset, \mathsf{a}, \mathsf{b}, \mathsf{ab}); \qquad \pi_0(\mathsf{ab}) = \frac{\beta^2}{3 + 2\beta + \beta^2} \leq \frac{\beta^2}{3}, \qquad 1 - \pi_0(\emptyset) = \frac{2\beta + \beta^2}{3 + 2\beta + \beta^2} \leq \beta,$$

the last inequality because $2\beta + \beta^2 \leq \beta(3 + 2\beta + \beta^2)$ for $\beta \geq 0$.

Kernels (identical for every $\varphi \in [0, 1]$, since γ_φ and d do not depend on φ).

- G_j: by (1) the inclusion probability of j is $\frac{\beta/6}{\beta/6+1/2} = \frac{\beta}{3+\beta}$ if the other coordinate is absent and $\frac{\beta^2/6}{\beta^2/6+\beta/6} = \frac{\beta}{1+\beta}$ if it is present; in every case it is $< \beta$. The other coordinate is unchanged.
- U: self-loop at $\emptyset$ and ab ($|S| \in \{0, p\}$). At a: delete 1, add 2, accept with $1 \wedge \gamma(\mathsf{b})/\gamma(\mathsf{a}) = 1$; so $\mathsf{a} \to \mathsf{b}$ surely, and likewise $\mathsf{b} \to \mathsf{a}$.
- N: self-loop at $\emptyset$. At ab: the deleted j has $N(j) \setminus \mathsf{ab} = \emptyset$, so the move is rejected. At a: delete 1; $N(1) \setminus \mathsf{a} = \{2\}$; propose b; accept with $1 \wedge [\gamma(\mathsf{b})/\gamma(\mathsf{a})] \cdot |N(1) \setminus \mathsf{a}|/|N(2) \setminus \mathsf{b}| = 1$. So $\mathsf{a} \to \mathsf{b}$ surely, and likewise $\mathsf{b} \to \mathsf{a}$.
- I: the proposal $S' \sim \pi_0$ is accepted with probability $1 \wedge e^{\varphi[d(S') - d(S)]} = 1$, so the output is a π_0 draw independent of everything before.

In particular U and N act deterministically as $\emptyset \mapsto \emptyset$, $\mathsf{ab} \mapsto \mathsf{ab}$, $\mathsf{a} \leftrightarrow \mathsf{b}$: they preserve $|S|$ and never move into or out of ab.

Here $S = \mathsf{a}$, $k = 1 < K$, $j = 2$, $r_1 = \frac{2}{(4-2)\sqrt{n+1}} = (n+1)^{-1/2}$ and, since $\sqrt{n+1} = \sqrt{1+\beta^2}/\beta$,

$$h_1 = \frac{1}{1+\sqrt{n+1}} = \frac{\beta}{\beta+\sqrt{1+\beta^2}}. \tag{5}$$

All probabilities below are for one round started at $X_0 = \mathsf{a}$, with σ uniform on the $5! = 120$ orders.

Lemma 2 (Coupling)**.** *Let $K_1, \dots, K_m$ and $K'_1, \dots, K'_m$ be Markov kernels on a finite set with $\sup_x \|K_i(x,\cdot) - K'_i(x,\cdot)\|_{\mathrm{TV}} \le \delta$ for each i, where $\|\mu - \nu\|_{\mathrm{TV}} = \sup_E |\mu(E) - \nu(E)|$. Let P, P' be the laws of the paths $(X_0, \dots, X_m)$ with the same X_0 and $X_i \sim K_i(X_{i-1}, \cdot)$, resp. $K'_i(X_{i-1}, \cdot)$. Then $|P(E) - P'(E)| \le m\delta$ for every set E of paths.*

Proof. For $0 \le l \le m$ let $P^{(l)}$ be the path law that uses $K'_1, \dots, K'_l$ and then $K_{l+1}, \dots, K_m$; $P^{(0)} = P$ and $P^{(m)} = P'$. Under $P^{(l-1)}$ and $P^{(l)}$ the path $(X_0, \dots, X_{l-1})$ has the same law λ, and the kernels after step l are the same; let $F(x_{0:l}) \in [0,1]$ be the conditional probability of E given $X_{0:l} = x_{0:l}$ under these later kernels. Then $|P^{(l-1)}(E) - P^{(l)}(E)| = \big|\sum_{x_{0:l-1}} \lambda(x_{0:l-1}) \sum_{x_l} [K_l - K'_l](x_{l-1}, x_l) F(x_{0:l})\big| \le \delta$, because $|\sum_z (\mu-\nu)(z) f(z)| \le \|\mu - \nu\|_{\mathrm{TV}}$ for probability vectors μ, ν and f with values in $[0,1]$. Sum over l. □

Comparison chain. Define Markov kernels K^0 on the same four states for the same five labels directly (they are not the method at any β): $K^0_{G_j}(S,\cdot) = \delta_{S\setminus\{j\}}$ (j removed surely, the other coordinate unchanged); $K^0_U = K_U$ and $K^0_N = K_N$ (the deterministic maps above); $K^0_I(S,\cdot) = \delta_\emptyset$. Given σ this chain is deterministic, it never adds a variable, and $\emptyset$ is absorbing. At every state, the total-variation distance between a step of the actual chain and the same step of the comparison chain is at most β: for G_j it is the inclusion probability ($< \beta$), for U, N it is 0, and for I it is $1 - \pi_0(\emptyset) \le \beta$. By Lemma 2 with $m = 5$, applied for each fixed σ and then averaged over σ,

$$|\mathbb{P}(E) - \mathbb{P}^0(E)| \le 5\beta \qquad \text{for every event } E \text{ of } (\sigma, X_0, \dots, X_5). \tag{6}$$

Lemma 3. *Let E_2 be the event that the state just before step G_2 is a, and E_1 the event that the state just before step G_1 is b. Then $\mathbb{P}^0(E_2) = \frac{30}{120}$ and $\mathbb{P}^0(E_1) = \frac{16}{120}$.*

Proof. In the comparison chain started at a: U and N swap $\mathsf{a} \leftrightarrow \mathsf{b}$ ("flips"); G_1 sends $\mathsf{a} \to \emptyset$ and fixes b; G_2 fixes a and sends $\mathsf{b} \to \emptyset$; I sends everything to $\emptyset$, which is absorbing. Let P be the set of labels before G_2 (orders written left to right). E_2 requires $I \notin P$ and an even number of flips in P. If $G_1 \in P$, G_1 must act at b, so exactly one flip precedes it and, by parity, one follows it before G_2. The possibilities are: $P = \emptyset$ (G_2 first: $4! = 24$ orders); $P = \{U, N\}$ (2 orders of U, N, then G_2, then 2 orders of G_1, I: 4 orders); $P = \{G_1, U, N\}$ in the order UG_1N or NG_1U, then G_2, then I: 2 orders. Total 30. E_1 requires I not before G_1 and an odd number of flips before G_1, so exactly one; and G_2, if before G_1, must act at a, i.e. before that flip. The possibilities are: $P' = \{U\}$ or $\{N\}$ as the set before G_1 (UG_1 or NG_1 first, then the other three labels in $3! = 6$ orders: 12 orders); $P' = \{G_2, U\}$ or $\{G_2, N\}$ in the order G_2, flip, G_1, then the remaining two labels in 2 orders: 4 orders. Total 16. □

Proposition 4 ((ii-neg): no ordering-free bound)**.** *In the example, for every $n \ge 400$ and every $\varphi \in [0,1]$,*

$$\mathbb{P}(\textit{visit } \mathsf{ab} \mid X_0 = \mathsf{a}) \le \tfrac{23}{60}\beta + \tfrac{31}{3}\beta^2 < h_1.$$

So the inequality $\mathbb{P}(\textit{visit } S + j \mid X_0 = S) \ge h_k$ fails at $S = \mathsf{a}$, $j = 2$, $k = 1$.

Proof. If ab is visited, let $i \geq 1$ be the first step with $X_i = \mathsf{ab}$ ($X_0 = \mathsf{a} \neq \mathsf{ab}$). Step $\sigma(i)$ is not U or N, since they never move into ab. If it is G_2, then X_{i-1} agrees with ab off coordinate 2 and $X_{i-1} \neq \mathsf{ab}$, so $X_{i-1} = \mathsf{a}$; if it is G_1, then $X_{i-1} = \mathsf{b}$; otherwise it is I. Hence

$$\mathbb{P}(\text{visit } \mathsf{ab}) \leq \mathbb{P}(I \text{ outputs } \mathsf{ab}) + \mathbb{P}(E_2 \text{ and } G_2 \text{ adds } 2) + \mathbb{P}(E_1 \text{ and } G_1 \text{ adds } 1).$$

The first term is $\pi_0(\mathsf{ab}) \leq \beta^2/3$. E_2 is determined by σ and the path before G_2, and the G_2 draw uses fresh randomness, so $\mathbb{P}(E_2 \text{ and } G_2 \text{ adds}) = \mathbb{P}(E_2)\, o_2(\mathsf{a}, \varphi) = \mathbb{P}(E_2)\frac{\beta}{1+\beta}$; likewise $\mathbb{P}(E_1 \text{ and } G_1 \text{ adds}) = \mathbb{P}(E_1)\frac{\beta}{1+\beta}$. By (6) and Lemma 3, $\mathbb{P}(E_1) + \mathbb{P}(E_2) \leq \frac{46}{120} + 10\beta = \frac{23}{60} + 10\beta$. Therefore

$$\mathbb{P}(\text{visit } \mathsf{ab}) \leq \frac{\beta^2}{3} + \beta\Big(\frac{23}{60} + 10\beta\Big) = \frac{23}{60}\beta + \frac{31}{3}\beta^2.$$

By (5), this is $< h_1$ iff $f(\beta) := \big(\frac{23}{60} + \frac{31}{3}\beta\big)\big(\beta + \sqrt{1+\beta^2}\big) < 1$. f is increasing on $(0, \infty)$, and $f(1/20) = \frac{54}{60} \cdot \frac{1+\sqrt{401}}{20} < 0.9 \times 1.0513 < 0.95$. Since $n \geq 400$ means $\beta \leq 1/20$, the claim follows. □

Proposition 5 ((iv-a): terminal occupancy; second display supplementary)**.** *In the example, for every $n \geq 400$ and every $\varphi \in [0, 1]$,*

$$\mathbb{P}(X_5 = \mathsf{ab} \mid X_0 = \mathsf{a}) \leq \tfrac{4}{3}\beta^2 < \binom{5}{1}^{-1} h_1, \qquad \frac{\mathbb{P}(X_5 = \mathsf{ab} \mid X_0 = \mathsf{a})}{\binom{5}{1}^{-1} h_1} \leq \tfrac{20}{3}\beta\big(\beta + \sqrt{1+\beta^2}\big) \xrightarrow[n\to\infty]{} 0.$$

Proof. Let G° be the earlier and G^* the later of the two Gibbs steps in σ, and j^* the coordinate of G^*. Since U, N preserve $\{\mathsf{ab}\}$ and its complement, and all steps after G^* other than possibly I are among U, N, we split on the position of I (these three cases partition the orders, and σ is independent of the chain's draws). *I after G^*.* After I the state is a fresh π_0 draw and every later step is U or N; so $X_5 = \mathsf{ab}$ has conditional probability $\pi_0(\mathsf{ab}) \leq \beta^2/3$. *$I$ between G° and G^*.* After I the state is a π_0 draw and the steps up to G^* are among U, N, which preserve size; so the state before G^* is nonempty with conditional probability $1 - \pi_0(\emptyset) \leq \beta$. All steps after G^* are among U, N, so $X_5 = \mathsf{ab}$ forces the state right after G^* to be ab, which requires a nonempty state before G^* and the inclusion of j^* at G^*; the latter has conditional probability $< \beta$ given the past. So the conditional probability is $\leq \beta^2$. *I before G°.* As in the previous case, the state before G° is nonempty with probability $\leq \beta$; if it is empty, G° makes it nonempty with probability $\frac{\beta}{3+\beta} \leq \frac{\beta}{3}$. The steps between G° and G^* are among U, N. So the state before G^* is nonempty with probability $\leq \frac{4}{3}\beta$, and as before the conditional probability of $X_5 = \mathsf{ab}$ is $\leq \frac{4}{3}\beta^2$. Each case has conditional probability $\leq \frac{4}{3}\beta^2$, hence so does the mixture. By (5), $\binom{5}{1}^{-1} h_1 = \beta/\big(5(\beta + \sqrt{1+\beta^2})\big)$, which gives the ratio bound. The ratio bound is increasing in β, is $\leq \frac{20}{3} \cdot \frac{1}{20} \cdot 1.0513 < 0.36 < 1$ at $\beta = 1/20$, and tends to 0 as $\beta \to 0$. □

Remark 6. (a) The example has $p = 2$. Here $\sum_{k \leq K} \binom{2}{k} = 4 \leq (5H_0 + 4N_0) \cdot 2$, so by the exact-branch rule quoted in STATEMENT.md the production method enumerates all models and runs no SMC round. (ii-neg) and (iv-a) are statements about the round kernel K_φ, which is defined for all admissible data, and the example is admissible; but the example does not exhibit the failure in a regime where production SMC rounds are run. (b) The visit probability in the example is $\geq h_1/5$, as part (ii) requires. (c) Corroboration, not used in the proofs above. In the example $d \equiv 0$, so $q_\varphi = \pi_0$; every elementary kernel is π_0-reversible (T09(ii)), hence π_0-invariant; the I step outputs an exact π_0 draw; and the order is independent of the draws. So the state after I has law π_0 and keeps it, and $X_5 \sim \pi_0$ exactly: $\mathbb{P}(X_5 = \mathsf{ab} \mid X_0 = \mathsf{a}) = \pi_0(\mathsf{ab}) = \beta^2/(3 + 2\beta + \beta^2)$, consistent with Proposition 5. Exact rational enumeration of all 120 orders at $n = 400$ gives visit probability $0.01978 < h_1 = 0.04756$ and terminal occupancy $8.06 \cdot 10^{-4} = \pi_0(\mathsf{ab}) < h_1/5 = 0.00951$.

5 Part (iii): execution-time hazards

Model of a run (EX-A, EX-S). Enumerate all elementary steps of a run (all particles, rounds and stages, of the pilot, the calibration and the final run) as $u = 1, 2, \dots$, in an order in which they can be executed sequentially (within a round, particles one after another; resampling and adaptive choices between steps). (The step index is u; e is reserved for the budget tier.) For u beyond the last step put $\lambda_u = \text{none}$. Let $\mathcal{F}_{u^-}$ be the σ-field generated by the data and by every stream variate consumed before the own draw of step u (including the round orders, resampling and adaptive choices made so far). It determines the label λ_u, the temperature φ_u used by step u and the current state X_{u^-} of the particle executing it; X_u is that particle's state after step u. By the method (STATEMENT.md, "Temperatures"), $\varphi_u \in [0,1]$ for every move; this is the hypothesis of (iii). Under EX-A and EX-S, step u draws X_u from the elementary kernel of label λ_u at temperature φ_u at the current state, using the next stream variate(s) not yet consumed. Under EX-S the stream is i.i.d. uniform; the number of variates consumed before the draw of step u is a stopping time for the stream filtration (whether it equals t is decided by the first t variates and the data), and $\mathcal{F}_{u^-}$ is the corresponding stopped σ-field; by the strong Markov property of an i.i.d. sequence at a stopping time, the next variates are uniform and independent of $\mathcal{F}_{u^-}$. Hence, in particular, on $\{\lambda_u = G_j\}$,

$$\mathbb{P}(X_u = \cdot \mid \mathcal{F}_{u^-}) = K_{G,j}(X_{u^-}, \cdot)\big|_{\varphi=\varphi_u} \quad \text{a.s.} \tag{7}$$

(iii-a). Fix $S \in \Omega$ with $|S| = k < K$ and $j \notin S$, and let $A_u = \{\lambda_u = G_j,\ X_{u^-} = S\} \in \mathcal{F}_{u^-}$. On A_u, (7) and (1) give $\mathbb{P}(X_u = S + j \mid \mathcal{F}_{u^-}) = o_j(S, \varphi_u)$, and $o_j(S, \varphi_u) \geq h_k$ by part (i), because $\varphi_u \in [0,1]$ and h_k does not depend on φ.

(iii-b). The map $\varphi \mapsto o_j(S, \varphi)$ is continuous, so $o_j(S, \varphi_u)\mathbf{1}_{A_u}$ is $\mathcal{F}_{u^-}$-measurable. By Tonelli's theorem (nonnegative terms) and the tower property,

$$\mathbb{E}N_j(S) = \sum_u \mathbb{E}\big[\mathbf{1}_{A_u}\mathbf{1}\{X_u = S{+}j\}\big] = \sum_u \mathbb{E}\big[\mathbf{1}_{A_u}\mathbb{P}(X_u = S{+}j \mid \mathcal{F}_{u^-})\big] = \sum_u \mathbb{E}\big[\mathbf{1}_{A_u}\, o_j(S, \varphi_u)\big] \geq h_k \sum_u \mathbb{P}(A_u),$$

with values in $[0, \infty]$.

Conditional form. Let $\mathcal{G}$ be a σ-field and $\mathcal{I}$ a (possibly random) set of step indices such that, for every u: $\{u \in \mathcal{I}\} \in \mathcal{G}$; $B \cap \{u \in \mathcal{I}\} \in \mathcal{F}_{u^-}$ for every $B \in \mathcal{G}$; and $\varphi_u \mathbf{1}\{u \in \mathcal{I}\}$ is $\mathcal{G}$-measurable. Fix u and $B \in \mathcal{G}$, and write $I_u = \mathbf{1}\{u \in \mathcal{I}\}$. Since $B \cap \{u \in \mathcal{I}\} \cap A_u \in \mathcal{F}_{u^-}$, (iii-a) gives $\mathbb{E}[\mathbf{1}_B I_u \mathbf{1}_{A_u}\mathbf{1}\{X_u = S + j\}] = \mathbb{E}[\mathbf{1}_B I_u \mathbf{1}_{A_u} o_j(S, \varphi_u)] = \mathbb{E}[\mathbf{1}_B I_u\, o_j(S, \varphi_u)\mathbb{P}(A_u \mid \mathcal{G})]$, the last step because $I_u\, o_j(S, \varphi_u)$ is $\mathcal{G}$-measurable. Hence $\mathbb{E}[I_u \mathbf{1}_{A_u}\mathbf{1}\{X_u = S + j\} \mid \mathcal{G}] = I_u\, o_j(S, \varphi_u)\mathbb{P}(A_u \mid \mathcal{G})$ a.s., and summing over u (conditional monotone convergence) gives, almost surely,

$$\mathbb{E}\Big[\sum_{u \in \mathcal{I}} \mathbf{1}_{A_u}\mathbf{1}\{X_u = S + j\} \,\Big|\, \mathcal{G}\Big] = \sum_{u \in \mathcal{I}} \mathbb{P}(A_u \mid \mathcal{G})\, o_j(S, \varphi_u) \ \geq\ h_k \sum_{u \in \mathcal{I}} \mathbb{P}(A_u \mid \mathcal{G}).$$

Illustration (method facts from 00_COMMON §2, not used elsewhere): $\mathcal{G}$ generated by the data and all variates consumed before the final run starts (so $\mathcal{G} \supseteq \sigma(\mathcal{C})$), and $\mathcal{I}$ the final-run steps enumerated stage by stage, particle by particle and position by position; the final run starts after $\mathcal{G}$ is realised and its temperatures are fixed by the frozen grid, so the three conditions hold. With $\mathcal{G}$ trivial and non-random φ_u this is the displayed form of the theorem, the sum running over the executions of coordinate j ($A_u = \emptyset$ when $\lambda_u \neq G_j$). □

6 Part (iv)

(iv-a) is Proposition 5; the supplementary ratio statement is its second display.

(iv-b). Let $0 \le i < a$, so $1 \le k+i+1 \le K \le p$. Then $r_{k+i} = \frac{k+i+1}{(2p-k-i-1)\sqrt{n+1}}$ with $p \le 2p-k-i-1 \le 2p$, so $\frac{1}{2p\sqrt{n+1}} \le r_{k+i} \le \frac{1}{\sqrt{n+1}} < 1$, hence $h_{k+i} = \frac{r_{k+i}}{1+r_{k+i}} \ge \frac{r_{k+i}}{2} \ge \frac{1}{4p\sqrt{n+1}}$. Also $B \le \lfloor \sqrt{p} \rfloor \le p$, so $m = p+B+2 \le 2p+2 \le 4p$, and $\binom{m}{a} \le m^a/a!$. Therefore

$$\binom{m}{a}^{-1} \prod_{i<a} h_{k+i} \ge \frac{a!}{(4p)^a} \cdot \frac{1}{(4p\sqrt{n+1})^a} = a! \left(16p^2\sqrt{n+1}\right)^{-a} \ge a! \left(16\sqrt{2}\, n^{2A+1/2}\right)^{-a},$$

using $p \le n^A$ and $n+1 \le 2n$ in the last step. By part (ii), $\mathbb{P}(\text{visit } S \cup J \text{ during the round} \mid X_0 = S) \ge a!\,(16\sqrt{2}\, n^{2A+1/2})^{-a}$. This is uniform over the data, S, J and $\varphi \in [0,1]$, and inverse-polynomial in n for fixed a. □

(iv-c). Scope: the final run with the declared budgets of 00_COMMON §2, either tier $e \in \{0,1\}$. Let $a = 1$, k fixed, $p = n$ and $n \ge \max(2500, 2k+2)$, so that $k < K = \lfloor n/2 \rfloor$. Using $2p-k-1 \ge p$,

$$h_k \le r_k \le \frac{k+1}{p\sqrt{n}} = (k+1)n^{-3/2}, \qquad \frac{h_k}{m} \le \frac{r_k}{p} \le (k+1)n^{-5/2} \quad (m \ge p).$$

By the budget block and final-run structure quoted in STATEMENT.md from 00_COMMON §2, the final run has $4N_e$ particles (4 islands $\times N_e$) and $T+L+4$ rounds per particle (T grid stages with one round each, L burn-in rounds and 4 measured rounds). Moreover $T \le T_e$, from $Q_f = 4N_e(T+L+5) \le 4N_e(T_e+L+5)$; $N_e = 4^e N_0 = 4^e \max(2500, n, p) = 4^e n \le 4n$ (as $p = n \ge 2500$); $\ell = \lceil \log_2(n+p+1) \rceil = \lceil \log_2(2n+1) \rceil$; $T_e = 4^e T_0 = 4^e \max(1000, \ell^2) \le 4(1000+\ell^2)$; and $L = \max(4, \lceil \log_2(n+1) \rceil) \le \ell + 4$. So the number of final-run particle-rounds is

$$R = 4N_e(T+L+4) \le 16n\,(4008 + 4\ell^2 + \ell).$$

(1) The (ii) bound summed over the final run: $R \cdot h_k/m \le 16(k+1)\, n^{-3/2}(4008+4\ell^2+\ell) \to 0$. (2) Each round contains exactly one Gibbs-j step per particle, so the final run has exactly R Gibbs-j executions, and the (iii) bound satisfies $h_k \sum_{u \in \text{final run}} \mathbb{P}(A_u) \le h_k R \le 16(k+1)\, n^{-1/2}(4008+4\ell^2+\ell) \to 0$. Both limits hold because $\ell = O(\log n)$. So neither (ii) nor (iii), summed over the final-run budget, yields a quantity bounded away from 0; no discovery guarantee within that budget follows from them. The theorem asserts nothing about discovery. □

(iv-d). If $|S| = K$ and $j \notin S$, then $|S+j| = K+1$, so $S+j \notin \Omega$, $\varpi(S+j) = 0$, $\gamma_\varphi(S+j) = 0$, and by (1) (T09(ii)) $o_j(S,\varphi) = 0$: no Gibbs step adds a variable at a size-K state (the denominator $\gamma_\varphi(S^+) + \gamma_\varphi(S^-) \ge \gamma_\varphi(S) > 0$ by T22 Lemma 1). No other kernel leaves Ω: both swaps propose $S \setminus \{j\} \cup \{l\}$, of size $|S|$, and the independence proposal is a π_0 draw, supported on Ω. Parts (i)–(iii) are asserted, and proved above, only for $k < K$ and, in (ii), $k + a \le K$. □

Hypothesis use

Part (i) and (iv-b): deterministic, for every admissible data set; no statistical assumption. Part (ii): the round's construction (uniform order, independent of the state) and EX-S. Part (iii): EX-A and EX-S in the form (7), and $\varphi_u \in [0,1]$ for moves. T09(ii): the Gibbs kernel (1) (and, in Remark 6(c)

only, reversibility of all elementary kernels). T22 Lemmas 1–3: positivity, the score floor (2) and the prior ratio. 00_COMMON budget block and exact-branch rule (quoted in STATEMENT.md): (iv-c) and Remark 6(a). SR and CAL: not used.